

#MATLAB Assignment_2

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%% Question 1
m=[2, 4, 1; 6, 7, 2; 3, 5, 9];
m1=m([2 3 2 3],:);
%(a)Delete 2nd row
m1(2,:)=[];
%(b)Find sub-matrix of order 2-by-2
m2=m(1:2,1:2);

%% Question 2
A = [2, 4, 1; 6, 7, 2; 3, 5, 9]; %
B = [7, 5, 6; 2, 0, 8; 5, 7, 1]; %
Add=A+B;
Subtract=A-B;
%(a)matrix multiplication
Multiplication=A*B;
%(b) element-wise matrix multiplication
elementwise=A.*B;

%% Question 3
A = [2, 4, 1; 6, 7, 2; 3, 5, 9];
B = [7, 5, 6; 2, 0, 8; 5, 7, 1];
%the left division and right division of the matrix
right_division=A/B; %A*inv(B) / (slash)
left_division=A\B; %inv(A)*B \ (Backslash)

%% Question 4
A = [10, 12, 23; 14, 8, 6; 27, 8, 9]; % matrix
b = 5; %scalar
%the scalar operations on the matrix A are as following
add=A+b;
subtract=A-b;
multiply=A*b;
divide=A/b;

%% Question 5
A = [10, 12, 23; 14, 8, 6; 27, 8, 9];
B = [12, 31, 45; 8, 0, -9; 45, 2, 11];
horizontal_concatenation = horzcat(A,B); %horizontal
concatenation
vertical_concatenation = vertcat(A,B); %vertical concatenation

%% Question 6
%(a)
I=eye(3,3);%eye(3) %Identity matrix
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% (b)
O=zeros(3,3);%Zero matrix
% (c)
R=rand(3,3); %rand(3)%Random matrix
M=magic(3);%Magic matrix
```

```
%% Question 7
arr2=[1 2;5 6;3 4;7 8];%array 2D
arr1=[1 ;2;3;4]; %array 1D
Y=horzcat(arr2, arr1) ;
```

```
%% Question 8
% (a)
a= 3;
b= 4;
%Relational Operations
a==b;%equality
a>=b;%greater than or equal to
a>b; %greater than
a<=b;%less than or equal to
a<b;%less than
a~=b; %not equal
```

```
% (b)
a= 10;
b= 5;
%Relational Operations
a==b;%equality
a>=b;%greater than or equal to
a>b; %greater than
a<=b;%less than or equal to
a<b;%less than
a~=b; %not equal
```

```
%% Question 9
% (a)
a= 10;
b= 0;
%logical operators (AND,OR,NOT)
a&b;%AND
a|b;%OR
a=~(b); %NOT
```

```
% (b)
a= 2;
b= 5;
```

```
%logical operators (AND,OR,NOT)
a&b;%AND
a|b;%OR
a=~(b); %NOT
```

```
%% Question 10
%(a) 2+3==5 OR 3 >7
2+3==5|3>7
%(b) 4+5 == 3 AND 4 == 4
4+5==3&4==4
%(c) NOT 7 > 3
~(7)>3
```

```
%% Question 11
tic
x1=linspace(1,2);% or
x2=linspace(1,2,100);
toc
```

```
% a = input('Enter the number of rows');
% b = input('Enter the number of coloumns');
% for i=1:a
%     for j=1:b
%         p(i,j)=input('Enter the elements');
%     end
% end
% p=reshape(p,a,b)
```